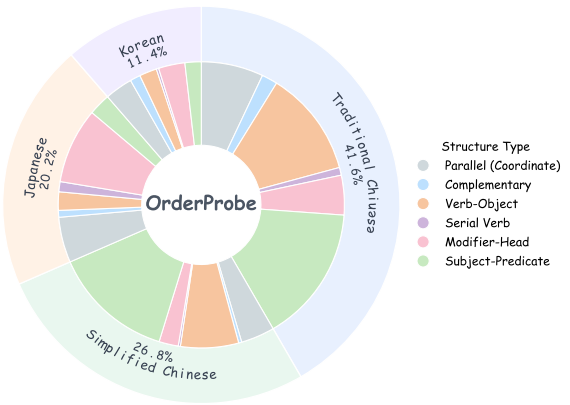

Figure 3: **Overview of OrderProbe.** The chart displays the distribution of script typologies and syntactic structures across the 3543 samples.

Category	Metric	Linguistic Definition & Rationale
Primary	**Recovery Rate**	Exact match of canonical order x. Measures global structural integrity.
Diagnostics	S_{Rob}	Robustness score based on degradation under S_n permutations (MDR/MDA).
	S_{Acc}^{mean}	Semantic fidelity via metaphorical alignment using Cross-Encoders and BERTScore.
	S_{Logic}	NLI-based entailment check between model output e and reference meaning.
	S_{Cons}	Stability across permutation variants, capturing invariance to internal shuffles.
	S_{Info}	Information density penalizing verbosity while rewarding high information-to-token ratio.

Table 2: **Evaluation Metrics on OrderProbe.** We prioritize Recovery Rate as the global structural indicator. Other diagnostic metrics characterize robustness, semantic fidelity, logical validity, invariance, and information density.

categories. The distribution of script typologies and structural patterns is shown in Figure 3. This design supports controlled analyses of how reconstruction varies across writing systems and syntactic templates. We additionally treat Korean Hangul as a phonogrammatic negative control: unlike logographic scripts, scrambling Hangul characters disrupts syllable composition and removes local semantic anchors, making deterministic reconstruction fundamentally harder.

3 Evaluation Framework

Beyond deterministic exact-match reconstruction, OrderProbe contributes a multi-dimensional diagnostic evaluation framework that disentangles structural planning from semantic recall. A single aggregate score is insufficient: models may fail recovery due to hallucinated retrieval, positional sensitivity, format mimicry, or verbose "knowledge dumping," all of which can mask the underlying failure mechanism. We therefore design six complementary metrics(Table 2) computed over the full permutation space, enabling fine-grained failure-mode analysis.

Evaluation Scope. For each canonical expression x, we evaluate model behavior under all 23 non-identity permutations $\mathcal{P}(x)$, yielding $3543 \times 23 = 81{,}489$ perturbed inputs. Each output is scored by exact reconstruction, semantic alignment, and logical entailment, and then aggregated across permutations to quantify sensitivity and invariance. This produces a six-dimensional diagnostic signature per model, rather than only a leaderboard rank.

3.1 Primary Metric: Recovery Rate

The Recovery Rate is the global indicator of structural integrity:

$$\text{Recovery} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(\hat{x}_i = x_i). \tag{4}$$

Unlike sentence restoration (ill-posed due to multiple valid reorderings), four-character expressions have a unique canonical form, enabling deterministic evaluation.

3.2 Diagnostic Metrics (Beyond Recovery)

We introduce five diagnostic dimensions to explain *why* recovery fails.

Semantic Fidelity (S_{Acc}^{mean}). We evaluate explanation quality using a tiered hybrid semantic metric integrating cross-encoder relevance, multilingual embedding similarity, and lexical safeguards against entity-level hallucination (Appendix A.2).

Logical Validity (S_{Logic}). To detect fluent but contradictory definitions, we compute entailment probability between generated and reference meanings using a multilingual NLI model:

$$S_{\text{Logic}} = P_{\text{NLI}}(e \Rightarrow r). \tag{5}$$

Structural Consistency (S_{Cons}). A key novelty is a permutation-level invariance score. For each permutation, we measure the gap between the model's potential best semantic score and its realized mean score, and aggregate both average deviation and worst-case sensitivity across $\mathcal{P}(x)$. We define:

$$S_{\text{Cons}} = (1 - E_{\text{perf}}) \cdot (1 - R_{\text{sens}}), \tag{6}$$

which penalizes both global instability and localized brittleness (Appendix A.3).

Model Family	Setting	Diagnostic Component Metrics					Main Metric
		S_{Acc}^{mean}	S_{Cons}	S_{Logic}	S_{Rob}	S_{Info}	Recovery. (%)
Tier 1: High-Performance Reasoning Models							
Qwen-3-14B	Zero-shot	0.493	0.591	0.672	0.621	**0.579**	31.062
	+CoT	0.429	0.741	0.811	0.603	0.536	31.527 (↑0.465)
Qwen-3-VL$_{\text{Think}}$	Zero-shot	**0.520**	0.539	0.712	0.645	0.520	30.928
	+CoT	0.471	0.716	0.888	0.632	0.533	48.376 (↑17.448)
Gemini-2.5-Flash	Zero-shot	0.493	0.621	0.646	0.631	0.540	30.487
	+CoT	0.455	0.738	0.738	**0.647**	0.246	41.559 (↑11.072)
Tier 2: Mid-Range General Models							
GPT-4o	Zero-shot	0.463	0.663	0.591	0.616	0.489	23.254
	+CoT	0.431	0.769	0.774	0.622	0.400	37.687 (↑14.433)
GLM-4V$_{\text{Think}}$	Zero-shot	0.446	0.565	0.629	0.591	0.446	21.636
	+CoT	0.413	0.785	0.756	0.594	0.361	30.079 (↑8.443)
DeepSeek-V3.2	Zero-shot	0.472	0.593	0.575	0.609	0.444	20.104
	+CoT	0.482	0.727	**0.948**	0.643	**0.689**	**62.238** (↑42.134)
Tier 3: Efficient & Baseline Models							
DeepSeek-R1	Zero-shot	0.446	0.682	0.465	0.593	0.469	18.596
	+CoT	0.420	0.790	0.739	0.612	0.470	27.729 (↑9.133)
Claude-HaiKu-4.5	Zero-shot	0.448	0.656	0.494	0.612	0.466	18.017
	+CoT	0.398	0.714	0.652	0.551	0.293	21.646 (↑3.629)
Gemma-3-4B	Zero-shot	0.439	0.654	0.578	0.610	0.291	17.094
	+CoT	0.402	0.783	0.655	0.600	0.104	18.027 (↑0.933)
Llama-3-8B	Zero-shot	0.396	0.706	0.399	0.590	0.304	12.167
	+CoT	0.385	**0.815**	0.641	0.586	0.226	24.428 (↑12.261)
Llama-2-7B	Zero-shot	0.411	0.731	0.502	0.588	0.182	12.576
	+CoT	0.412	0.744	0.469	0.602	0.258	18.747 (↑6.171)
Qwen-3-8B	Zero-shot	0.446	0.617	0.487	0.604	0.403	15.786
	+CoT	0.392	0.715	0.492	0.580	0.269	12.594 (↓3.192)

Table 3: **Comparative Assessment of Zero Shot and Chain of Thought Prompting across 12 LLMs on OrderProbe.** This table reports performance metrics encompassing semantic accuracy, consistency and structural recovery.

Robustness (S_{Rob}). We quantify robustness as degradation relative to the canonical baseline and decompose it into sequential robustness (permutation sensitivity) and structural robustness (variance across syntactic categories), combined via harmonic mean (Appendix A.5).

Information Density (S_{Info}). To penalize verbosity-based metric inflation, we introduce an information density score using a brevity penalty that rewards informative explanations per token (Appendix A.7).

Takeaway. Together, these metrics expose systematic dissociations such as high semantic fidelity but low structural recovery, or high consistency due to rigid hallucination, enabling diagnosis beyond surface accuracy rankings. Full formulations and implementation details are provided in Appendix A.

4 Experiments

4.1 Experimental Setup

We evaluate all models on the full OrderProbe benchmark containing 3,543 canonical four-character expressions across four script settings (ZH-CN, ZH-TW, JA, KO). Given each scrambled input, models are required to output *(i)* the recovered canonical expression (a single four-character sequence) and *(ii)* a one-line semantic explanation in the corresponding language. All outputs are evaluated by exact-match reconstruction and our diagnostic metrics described in Section 3.

Models. We evaluate twelve widely used LLMs spanning open- and closed-source families. Open-source models include instruction-tuned checkpoints from Qwen (Qwen et al., 2025), DeepSeek (DeepSeek-AI et al., 2025), Llama (Grattafiori et al., 2024), and Gemma, run locally using official weights. Closed-source models include GPT

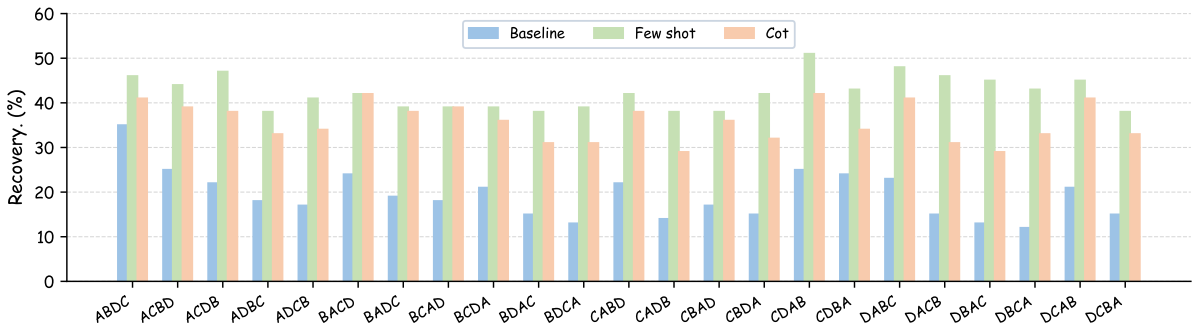

Figure 4: **Performance Comparison across 23 Permutation Patterns.** The chart illustrates recovery rates under baseline, few shot, and chain of thought settings, revealing the impact of anchor displacement on model stability.

Language	Component Metrics					Main Metric
	S_{Acc}^{mean}	S_{Cons}	S_{Logic}	S_{Rob}	S_{Info}	Recov. (%)
Simplified Chinese	**0.4817**	0.6445	**0.6854**	**0.6193**	0.4631	25.20
Traditional Chinese	0.4715	0.6184	0.6663	0.5998	0.4040	20.32
Japanese	0.4602	0.5897	0.5278	0.6048	**0.5388**	19.66
Korean	0.3990	**0.7625**	0.3014	0.5857	0.3906	5.72

Table 4: **Comparison across script settings.** Logographic scripts (Chinese/Japanese) yield substantially higher reconstruction accuracy than Korean Hangul, which serves as a phonogrammatic control condition. Component metrics are reported for diagnostic analysis alongside the primary Recovery score.

(OpenAI et al., 2024), Gemini, Claude (Anthropic, 2024), and GLM (Team et al., 2025), accessed via official APIs with provider-default settings during Oct 2025–Jan 2026. All checkpoint and deployment details are provided in Appendix A.10.

Prompting Strategy and Output Constraints. We use a unified two-message prompting format (system + user) with strict schema enforcement to minimize format drift. We adopt language-matched prompting: each script setting is evaluated using prompts written in the same language as the input, and models are instructed to produce both the recovered expression and explanation in that language. We compare three settings: **(i)** Zero-shot (one-line output), **(ii)** Chain-of-Thought (CoT; exactly two lines: reasoning + final answer), and **(iii)** three-shot in-context learning (few-shot; two-line output following exemplars). All prompt templates are provided in Appendix C.

4.2 Overall Results and Analysis

Table 3 summarizes the benchmark results across twelve models under both Zero-shot and CoT settings. Across model families, we observe a consistent semantic–structure dissociation: models can often generate fluent, meaning-aligned explanations while failing to reconstruct the canonical order under permutation noise.

Reconstruction remains challenging. Exact reconstruction is difficult even for strong systems. Under the zero-shot setting, recovery frequently remains below 35%, indicating that structural recovery is not an automatic byproduct of semantic competence. Instead, models tend to behave as if scrambled characters are independent tokens, producing semantically plausible but structurally incorrect outputs.

CoT helps, but gains vary widely. CoT prompting improves recovery for many models, suggesting that stepwise reasoning encourages explicit constraint checking before generation. However, improvements are highly model-dependent: some systems benefit substantially, while others show marginal gains or even regressions due to format drift and unstable decoding.

Consistency is not correctness. High structural consistency can occur even when recovery is low. In several cases, models respond with stable but generic explanations across permutations, producing high invariance scores without solving the reconstruction problem. This highlights the necessity of interpreting diagnostic metrics jointly rather than relying on a single dimension.

Summary. Overall, OrderProbe reveals that deterministic structural reconstruction remains a substantial capability gap. Models can retrieve meanings or provide plausible definitions, but often fail to enforce global ordering constraints required for exact canonical reconstruction.

4.3 Permutation Sensitivity and Prompting Effects

Figure 4 compares recovery trends across all 23 permutation configurations under baseline, CoT, and few-shot settings. We observe strong dependence on anchor displacement: permutations that preserve semantically informative characters near